

YOUNES NACEUR

ICFO - Universitat Politècnica de Catalunya - Université de Toulouse

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EDUCATION

ICFO - The Institute of Photonic Sciences , Barcelona, Spain	<i>October 2024 - Present</i>
Doctor of Philosophy, Physics	
Joint diploma from Universitat Politècnica de Catalunya and Université de Toulouse	
École Normale Supérieure , Paris, France	<i>September 2023 - July 2024</i>
Master of Science, Physics	GPA: 4.0/4.0
M2 International Center of Fundamental Physics (Theoretical Physics track)	
Sorbonne Université , Paris, France	<i>September 2022 - July 2023</i>
Master of Science, Physics	GPA: 4.0/4.0
M1 Paris Physics Master Programme, Best of the year	
Universidade de Lisboa , Lisbon, Portugal	<i>September 2020 - June 2021</i>
Exchange year with the Erasmus+ Programme	GPA: 4.0/4.0
Leibniz Universität Hannover , Hannover, Germany	<i>October 2018 - August 2022</i>
Bachelor of Science, Physics	GPA: 3.9/4.0
Minor in Computer Science	

HONORS AND SCHOLARSHIPS

Marie Skłodowska-Curie Doctoral Fellow	<i>January 2025</i>
issued by <i>European Commission</i>	
Full scholarship for a Master's programme abroad	<i>September 2022</i>
issued by <i>German Academic Exchange Service</i>	
Bachelor of Science with Honors	<i>August 2022</i>
<i>Leibniz Universität Hannover</i>	
Erasmus+ Scholarship	<i>September 2020</i>
<i>Universidade de Lisboa</i>	
Lower Saxony Scholarship	<i>October 2018</i>
issued by <i>Leibniz University Hannover</i>	
High school diploma prize and book prize	<i>June 2017</i>
issued by <i>German Physical Society</i>	

PROJECTS

Optimal pulse sequence design for dynamical decoupling in quantum control
<i>September 2026 - December 2026</i>
Industrial internship with the quantum computing company <i>Quandela</i> . Developed and certified optimal pulse sequences to improve coherent quantum control for Quandela's hybrid spin-photon quantum computing architecture.
Reachability and optimal-time certificates for quantum control
<i>January 2026 - June 2026</i>
Developed hierarchies of semidefinite relaxations to certify reachability and optimal-time limits in

constrained quantum control, yielding upper bounds on achievable state and gate fidelities and lower bounds on minimum control times under experimentally relevant control constraints [arXiv preprint](#).

Certified bounds on optimization problems in quantum theory *February 2025 - December 2025*

Developed methods to obtain exact lower bounds to non-commutative optimization problems from numerically inexact bounds as obtained through numerical solvers, providing a method to analytically certify bounds in quantum theory. [arXiv preprint](#).

Multi-Gaussian trajectory method for open quantum systems *April 2024 - July 2024*

Research internship at *The Materials and Quantum Phenomena Laboratory (MPQ)* about designing new numerical methods for the simulation of the dynamics of open bosonic quantum systems using Monte-carlo methods and parametrized Wigner functions in phase-space.

Simulation of Universal Blind Quantum Computation (UBQC) *April 2023 - July 2023*

Internship at the quantum communication company *VeriQloud*, development of a compiler for *Qiskit* quantum circuits into Universal Blind Quantum Computation protocols simulated using *SquidASM* with performance and security benchmarking on the simulation; experience in measurement-based quantum-computing (MBQC), UBQC, quantum network simulations. [Github repository](#)

Adiabatic quantum computing for optimization problems *October 2022 - November 2022*

Project in the framework of a lecture on numerical methods; simulated adiabatic quantum computation in Python for solving combinatorial optimization problems like number and graph partitioning

Quantum Search by measurement for optimization problems *April 2022 - August 2022*

Bachelor thesis completed summa cum laude; Graduated in the Quantum Valley Lower Saxony Framework: Simulation of Grover's quantum search algorithm on measurement based quantum computers; experience in quantum information theory, gate-based quantum computing, adiabatic quantum computing, tensor networks

TEACHING EXPERIENCE

Teaching Assistant for undergraduate Physics laboratories, *April 2022 - August 2022*
Leibniz Universität Hannover

Tutor for students in Mathematics and Physics, *May 2015 - May 2017*
Independent

FURTHER KNOWLEDGE

IT-Knowledge: L^AT_EX, Python (including packages Qiskit, Qutip, SquidASM, PyTorch, scikit-learn), C, Java, Julia, Mathematica, Matlab, Jupyter Notebook, Git, Microsoft Office, Logic Pro

Languages: German (Native), English (Fluent), Spanish (Conversational), French (Conversational)

International Experience: 8 Months of Work and Travel in Australia and South-East-Asia with work experience in several fields (2017-2018)